

Cartesian Vector

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

Magnitude

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Directions

$$\mathbf{u}_A = \frac{\mathbf{A}}{A} = \frac{A_x}{A} \mathbf{i} + \frac{A_y}{A} \mathbf{j} + \frac{A_z}{A} \mathbf{k}$$

$$\mathbf{u}_A = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}$$

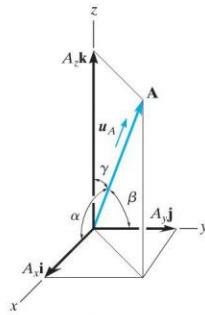
$$1 = \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$$

Cartesian Position Vector

$$\mathbf{r} = (x_2 - x_1) \mathbf{i} + (y_2 - y_1) \mathbf{j} + (z_2 - z_1) \mathbf{k}$$

Cartesian Force Vector

$$\mathbf{F} = F \mathbf{u} = F \left(\frac{\mathbf{r}}{r} \right)$$



Dot Product

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$$

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\theta = \cos^{-1} \left(\frac{\mathbf{A} \cdot \mathbf{B}}{AB} \right)$$

Moment of a Force

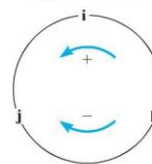
Scalar Definition

$$M_O = Fd$$

Vector Definition

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

Cross product sign convention



Moment of a Force About a Specified Axis

$$M_a = \mathbf{u}_a \cdot (\mathbf{r} \times \mathbf{F}) = \begin{vmatrix} u_{ax} & u_{ay} & u_{az} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

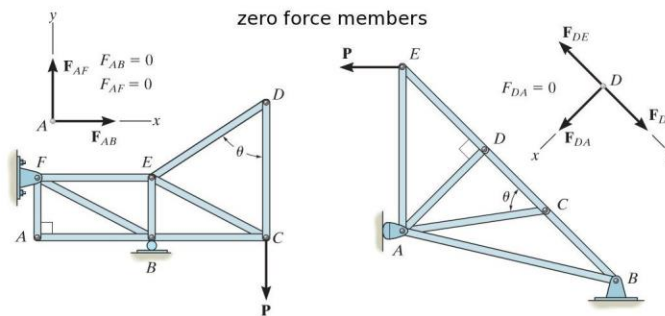
Simplification of a Force and Couple System

$$(F_R)_x = \Sigma F_x$$

$$(F_R)_y = \Sigma F_y$$

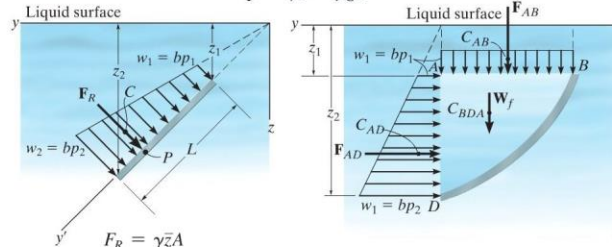
$$(M_R)_O = \Sigma M_O + \Sigma M$$

zero force members



Fluid Pressure

$$p = \gamma z = \rho g z$$



sign convention for internal forces / moments

Positive normal force



Positive moment



Positive shear



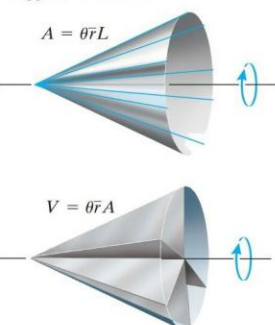
When drawing V and M diagrams

downward jump in M diag.

upward jump in M diag.



Pappus & Guldinus



Principal Moments and Moments of Inertia about an Inclined Axis

$$I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

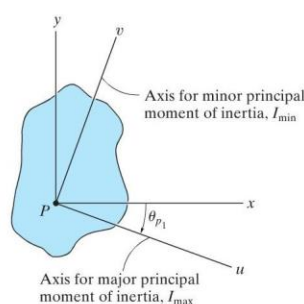
$$I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{uv} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

$$I_{u,v}^{\max} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2}$$

$$\tan 2\theta_p = \frac{-I_{xy}}{(I_x - I_y)/2}$$

Mohr's Circle



Equilibrium

Particle

$$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$$

Rigid Body-Two Dimensions

$$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M_O = 0$$

Rigid Body-Three Dimensions

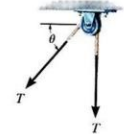
$$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$$

$$\Sigma M_x = 0, \Sigma M_y = 0, \Sigma M_z = 0$$

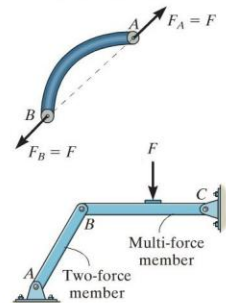
spring equation

$$F = ks$$

frictionless pulley cable is in tension



two force members

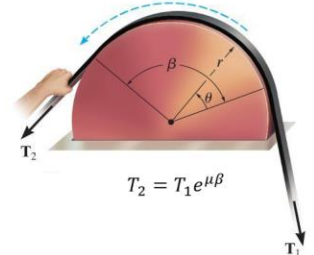


Friction

Static (maximum) $F_s = \mu_s N$

Kinetic $F_k = \mu_k N$

Motion or impending motion of belt relative to surface



Moments of Inertia

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

Product of Inertia

$$I_{xy} = \int_A xy dA$$

Parallel Axis Theorem

$$I = \bar{I} + Ad^2$$

$$I_{xy} = \bar{I}_{x'y'} + Ad_x d_y$$